

Problem Solving Strategies - Chapter 3

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1 Introduction

This might actually be one of the most important sections. I didn't fully understand the examples but hope that the problems will atleast teach me something.

2 Problems

2.1 Problem 1

Prove that there are at least $(2n - 2)/3$ triangles among the p_n parts of the plane in Example #1.

Solution

The first example is

(a) Into how many parts at most is a plane cut by n lines? (b) Into how many parts is space divided by n planes in general position?

p_n is the max number of lines. $p_n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2}$.

How can I uniquely determine a triangle? Could I determine it by the highest point at the top? Probably, I could always reorient the plane enough so that there is only one point.

What is needed to determine this max point? I would atleast 2 lines?

Wait, wait? What if I try to add interpretation to the equation? It seems like it is telling me that there are about $2n - 2$ edge point(vetices, or corners) and that only a 3 make up the triangle?

Well are there really $2n - 2$ intersection points? There should be about $\binom{n}{2}$ intersection points. Which comes out to be $n * (n - 1)/2$ points. I guess this is

Wait, this specifically specifies trianlges? I counted just inerseciton points, it could be the case that there are lots of quadrilaterals. But how do I guarantee that I get the triangel?

Which set of lines could make a triangle? I need an extremal property?

Actually let's make this interpretation, on each the lines there are 2 triangles, except one special line. Each traingle is counted 3 times, so we must divide by 3.

Now how do I show that 2 triangles rest on each line, pick a side, then choose the intersection point that is closest to the lines. This must make a triangle, because there is no intersection point that is closer.

So now I need to prove that there are max 2 lines that don't have an intersection on one side. It should be the case that if there are more than 2 then I can turn that into a triangle.

Lowk, am I cooked? I think I need to use an argument that it would violate the number of areas or somethigng but I don't know.

And the book doesn't say anything else. Guess I am cooked.

2.2 Problem 2

In the plane, n lines are given ($n \geq 3$), no two of them parallel. Through every intersection of two lines there passes at least an additional line. Prove that all lines pass through one point.

Solution

Assume that there it isn't the case that there is only one point of intersection. Then take the furthest intersection point to the left and then to the bottom.

2.3 Problem 5

Does there exist a tetrahedron, so that every edge is the side of an obtuse angle of a face?

Solution

I couldn't solve this one, but after taking a look at the answer it was so simple(just like all the problems).

So basically you take the longest edge in the tetrahedron, and you look at the look triangles it is a part of, there are 2. Since it is the longest, the side opposite the edge must be the greatest angle, therefore it can't be the case that this edge is an edge for an obtuse angle.

2.4 Problem 6

Prove that every convex polyhedron has at least two faces with the same number of sides.

Solution

This is just an application of pigeonhole principle, where you take the face with the maximal number of edges, call this number n . All the other faces must have between $[3, n]$. Since there at least n other faces there must at least be one 1 repetition.

2.5 Problem 7

$(2n + 1)$ persons are placed in the plane so that their mutual distances are different. Then everybody shoots his nearest neighbor.

Prove that

- (a) at least one person survives;
- (b) nobody is hit by more then five bullets;
- (c) the paths of the bullets do not cross;
- d) the set of segments formed by the bullet paths does not contain a closed polygon.

Solution

a) Take the two people, that are closest to each other, this people are point their guns at each other.

Then take the set of people that are not this, they can be grouped on whether they are pointing at one of this people.

Make S where S contains this two people that are point to each other, and the other people that are point to the 2 people. Define $f : S \mapsto S^*$, where $S^* = S \cup \{\text{people point their guns at people in } S\}$. Continually apply f to itself until the result stops growing. The last person added to this set will survive. Since now one is point their guns at him/her.

b) Take person who is hit by the most number of bullets. Then that means he is the person that is closest to the most number of people at the same time.

Take the person that is closest to him. Wait lowk, I think this is six bullets, since I can inscribe a perfect hexagon in a circle and everbody will have equal distance to each other. Actually after reading the mutual distance condition, then it is right. Also the book has a good angle argument, which I should have refined.

c) Yes, they don't cross because of the triangle inequality.

d) It can't because it can't contain a cycle. Should have explicitly shown the inequality.

My proof have no substance behind them. Fuck. Wait let there exist a closed polygon. Take the two closest people in this polygog, they are point their guns at each other. So now you have a chain, but it can't go back on itself.

2.6 Problem 8

Rooks are placed on the $n \times n$ chessboard satisfying the following condition:

If the square (i, j) is free, then at least n rooks are on the i^{th} row and j^{th} column together.

Show that there are at least $n^2/2$ rooks on the board.

Solution

Take the row/column that has the least number of rooks. Let this number be x , then there are at least $\max(n * x, n * (n - x))$, and the minimum of this expression is when $x = n/2$, therefore the minimum is $n^2/2$.

2.7 Problem 9

All plane sections of a solid are circles. Prove that the solid is a ball.

Solution

I don't know.

The shortest proof runs as follows. Consider the largest chord of the solid. Any section through this chord is a circle whose diameter is the chord. Otherwise the circle and the solid would have a larger chord. Thus the solid is a ball and one of its diameters is the selected chord. This proof is not complete. We did not prove that a longest chord exists. In fact, if the surface of the solid did not belong to the solid, a longest chord would not exist. So we assume that the solid is a closed and bounded set. Then we can apply the theorem of Weierstrass: A continuous function defined on a closed and bounded set always assumes its global maximum and minimum. This theorem belongs to higher mathematics, but at the IMO you can use it. The proof is not considered to have a gap if you cite the theorem. There are also elementary proofs which are slightly longer (see HMO 1954).

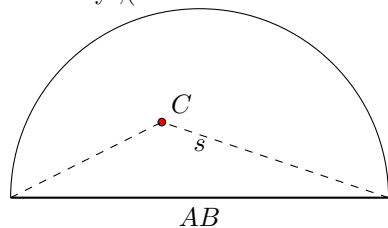
2.8 Problem 10

A closed and bounded figure ϕ with the following property is given in a plane: Any two points of ϕ can be connected by a half circle lying completely in ϕ . Find the figure ϕ (West German proposal for IMO 1977).

Solution

I don't know but after looking at the solution from the book, I realize it is maybe to geometry flavour for me, so I have listed the full solution for the sake of completeness below.

Too lazy ;(



2.9 Problem 11

Of n points in space, no four lie in a plane. Some of the points are connected by lines. We get a graph G with k edges.

- (a) If G does not contain a triangle, then $k \leq \lfloor n^2/4 \rfloor$.
- (b) If G does not contain a tetrahedron, then $k \leq \lfloor n^2/3 \rfloor$.

Solution

Where do I even start? What does Triangle have to with edges? What does the 4 points not in a plane thing have to do with anything? Well if there are 4 points in a plane, then There are 6 edges between them? What if they not in a plane then? Actually you can always make a triangle with 3 or more points, so the condition is basically saying that the points are not coplanar.

What do I do with this information? How does this affect anything though? What is the natural upper bound on the number of distinct edges? Well that would be $\binom{n}{2}$.

Holly shit, I reread the quesiton, I choose some subset of edges from the $\binom{n}{2}$, edges, and I call that Graph G . Now this graph doesn't have a triangel? So that are not cycles of length 3. That is what is it saying?

So what is the maximum of $E(G)$ for a graph G with n nodes such that it doesn't have a cycle of length 3.